

Welcome to Deep Learning for MSc

Course Overview

Hopefully you have read the course prerequisites on the Moodle Page?

DR KEVIN BRYSON AND DR ALI GOOYA

Course Lecturers



Dr Kevin Bryson
Room S113, SAWB

**Part 1: Week 1 – 5
Deep Learning
Fundamentals**

Office Hour(s):
3pm – 5pm
Tuesday



Dr Ali Gooya

**Part 2: Week 6 – 10
Deep Learning
Advanced Topics**

Who am I?

- [Dr Kevin Bryson](#), Senior Lecturer in Bioinformatics & AI
- Head of the [Information, Data, Analysis \(IDA\) Section](#)
- S113 SAWB, 18 Lilybank Gardens, SoCS

Co-Lead: [AI4BioMed Lab](#)

Member: [Inference, Dynamics & Interaction Group \(IDI\)](#)

Member: [Knowledge & Data Engineering Systems Group \(KDES\)](#)

Research Areas:

- AI in Biomedicine / Bioinformatics
- Distributed Agentic AI Systems
- Federated Learning Systems

Deep Learning for MSc Course Objectives

The aim of this course is to go beyond our introductory machine learning course and teach students about modern techniques for machine learning with high-dimensional image and sequence (time-series) data, and the underlying computational structures for such systems. Teach the students about managing large data sets, and the engineering pipelines for large-scale machine learning tasks.

By the end of this course students will be able to:

- Understand the major technology trends in advanced machine learning
- Build, train and apply fully connected deep neural networks
- Know how to implement efficient, vectorised neural networks in python and understand the underlying backends
- Apply deep learning methods to new applications
- Understand the machine learning pipeline, and engineering aspects of training data collation, and the importance of unlabelled data.

Lectures & Labs

Lectures are on Tuesday 2pm-3pm and 5pm to 6pm.

- These will mostly cover the theory behind the different deep learning topics to give you an understanding of how a variety of deep learning systems work. (I will sometimes relate the theory to PyTorch code though – but it is mostly trying to understand the concepts of DNNs.)

Your timetable should also show a 1-hour deep learning lab session on Thursday.

- These will be used to learn **practical skills of developing PyTorch deep learning models.**
- The lab exercises will typically take longer than 1 hour – ideally you would start doing these earlier than the lab and you will use the lab to ask questions and discuss different aspects with other students (discussion in the labs is encouraged).

Assessment to cover both theory and practice

There will be a Moodle quiz at the end of Part 1 (worth 10%) taken in the lab to consolidate your understanding of the **fundamental theory of the lectures** before you start advanced topics in Part 2.

Also, there will be a “lab report” (worth 16%) consisting of a submitted Jupyter Notebook which will check that you understand how to write PyTorch Deep Learning Models by the end of Part 1 of the labs – before moving onto more complex PyTorch code.

The key idea is not to just replicate “tutorial code” and say that you can do deep learning – it is to give you the fundamental theory and skills to know what this code is doing and develop novel code for your own deep learning models.

Assessment Summary

A Moodle Quiz assessing Part 1 theory (10%)

A Jupyter Notebook submission to assess PyTorch coding skills (16%)

Coursework that will get you to develop a full PyTorch workflow (14%)

Final exam paper in April / May (60%) – **resit available**.

The idea is to continually encourage you to engage with both the theory and practical material so that you are prepared for the final exam paper.

NOTE THAT YOU MUST COMPLETE 75% OF ALL COURSEWORK ON ANY COURSE TO BE COMPLETE FOR THAT COURSE – NOTE THE DANGER ON THIS COURSE !

For the course we will be using ... PyTorch

PyTorch's main features are:

- Efficient vectorized **tensor** operations on CPU and GPU
- Automatic on-the-fly differentiation (Autograd)
- Efficient data I/O for lots of different types of data (and built-in example data)
- Large variety of pre-built types of deep learning layers and models
- Lots of different types of neural network optimizers

“Efficient tensor operations” encompass both standard linear algebra and, as we will see later, and deep-learning specific operations (convolution, pooling, etc.)

Course Textbooks

Acknowledgement for course material and background reading from:

Understanding Deep Learning by Simon Prince. (Simon says the title of his book is a joke ...)

More mathematically deep (although older) is:

Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville.

A useful book for picking up PyTorch is:

Deep Learning with PyTorch by Eli Stevens and Luca Antiga.

(There are free versions of these books either online or via the UofG Library)

Part 1 Course Overview Weeks 1-5 (Fundamentals)

Week 1 – Introduction to deep learning

Lecture: historical overview, motivational examples, basic idea of a deep neural network

Lab: Quick overview introduction to PyTorch and practice with PyTorch Tensors.

Week 2 – Optimization of simple fully-connected neural network models

Lecture: Mathematical theory of fully-connected models.

Lab: PyTorch code for fully-connected models.

Week 3 – Convolutional neural networks for image analysis

Week 4 – Important aspects when constructing deep neural network pipelines

Week 5 – Optimization and Regularization in Deep Learning

Part 2 Course Overview Weeks 6-10

(More Advanced Topics – some typical topics)

Week 6 – Visualization and interpretation of deep neural networks

Week 7 – Learning from sequences (recurrent neural networks, transformers, etc.)

Week 8 – Unsupervised neural networks (Autoencoders) and Self-supervised (Barlow Twins).

Week 9 – Deep Generative Learning (Variational Autoencoders and GANs)

Week 10 – Ethics

(Sometimes Deep Reinforcement Models, Diffusions Models, etc.)

Deep Learning for MSc

Lecture 1 - Introduction

KEVIN BRYSON

The road to deep learning has been a long one

- Artificial neurons (McCullough-Pitts, 1943)
- Perceptron learning machine (Rosenblatt, 1958)
- Human visual cortex (Hubel-Wiesel, 1959)
- Neocognitron (Fukushima, 1980)
- Back-propagation to train ANNs (Rumelhart, Hinton, Williams, 1986)
- LeNet (first CNN) (Yann LeCun et al., 1989)
- **AlexNet 2012 (Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton)**
- Atari Game Playing using Deep Reinforcement Learning (2013)
- Generative Adversarial Networks (GANs) presented (Goodfellow et al. 2014)
- AlphaGo beats world champion (2016)
- GoogleBrain introduces the transformer model with “Attention is All You Need” (2017)
- Turing Award given to Yoshua Bengio, Geoffrey Hinton, Yann LeCun (2018)
- AlphaFold beats all the experts at protein folding in CASP13 (2019)
- OpenAI releases the large language model ChatGPT (2022)
- John J. Hopfield and Geoff Hinton got Nobel Prizes for Deep Neural Networks (2024)

Why did Deep Learning become successful ?

A combination of incremental improvements in:

- ❑ **Theory:** Five decades of improvements in ML theory.
- ❑ **Algorithms:** Five decades of improvements in algorithms.
- ❑ **Hardware:** CPUs/GPUs/storage developed for other purposes used.
- ❑ **Tools:** tools and culture of collaborative and reproducible science.
- ❑ **Data:** lots of data readily from “the Internet”, social networks, etc.
- ❑ **Money:** Resources and efforts from large corporations.
Thousands of researchers and developers attracted to the area.

New developments in *ML Theory*

Five decades of improvements in ML Theory:

- ❑ A taxonomy of ML concepts (supervised / unsupervised / generative models, kernels, embeddings, etc.),
- ❑ A sound statistical formalization (e.g. Bayesian estimation),
- ❑ A clear picture of fundamental issues and statistical modelling theory (bias/variance dilemma, VC dimension, generalization bounds, etc.)
- ❑ A theoretical understanding of shallow and deep networks (i.e. Cybenko's theorem).
- ❑ A good understanding of optimization issues (these are still not understood for deep learning!)
- ❑ Development of efficient large-scale vectorized algorithms.

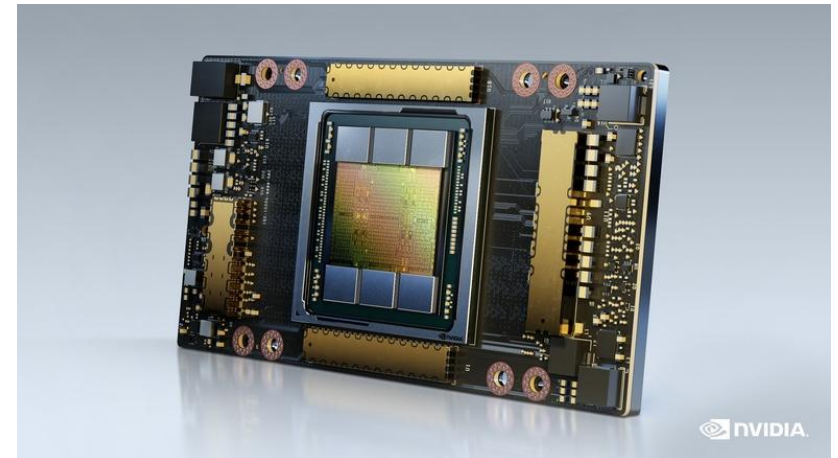
New developments in *Algorithms*

- ❑ Convolutional networks and Pooling methods
- ❑ Regularisation methods (e.g. Batch normalisation)
- ❑ Specific efficient activation functions (e.g. ReLU)
- ❑ Better weight initialisation techniques
- ❑ Improvements in optimisation techniques (e.g. Adam, etc.)

Once this collection of developments began to allow networks with more than 10 layers, we started to see deep learning improve its relative performance. Now networks of thousands of layers can be trained.

New developments in *Hardware*

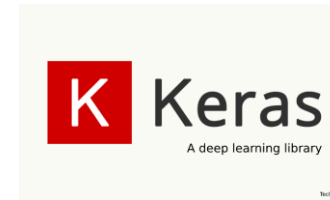
- CPUs in 2010 were 5000 times faster than 1990
- GPUs development was driven by graphics and the games industry
- NVIDIA then launched CUDA in 2007 – a programming interface for their GPUs.
 - GPUs give factor 10* improvement over powerful CPUs
- NVIDIA saw the ML market and developed specific hardware such as A100 with more memory and specialist ML cores



New developments in *Tools* (and culture of collaborative and reproducible science)

Better open-source development environments

- TensorFlow, Theano
- Python, NumPy
- Keras
- PyTorch



Faster sharing of scientific papers via **arXiv** pre-print server.

Peer pressure on authors to share working implementations and data.

Open, competitive environments such as **Kaggle** can allow engineers to push the envelope, and companies can gauge their performance against a wider range of participants.

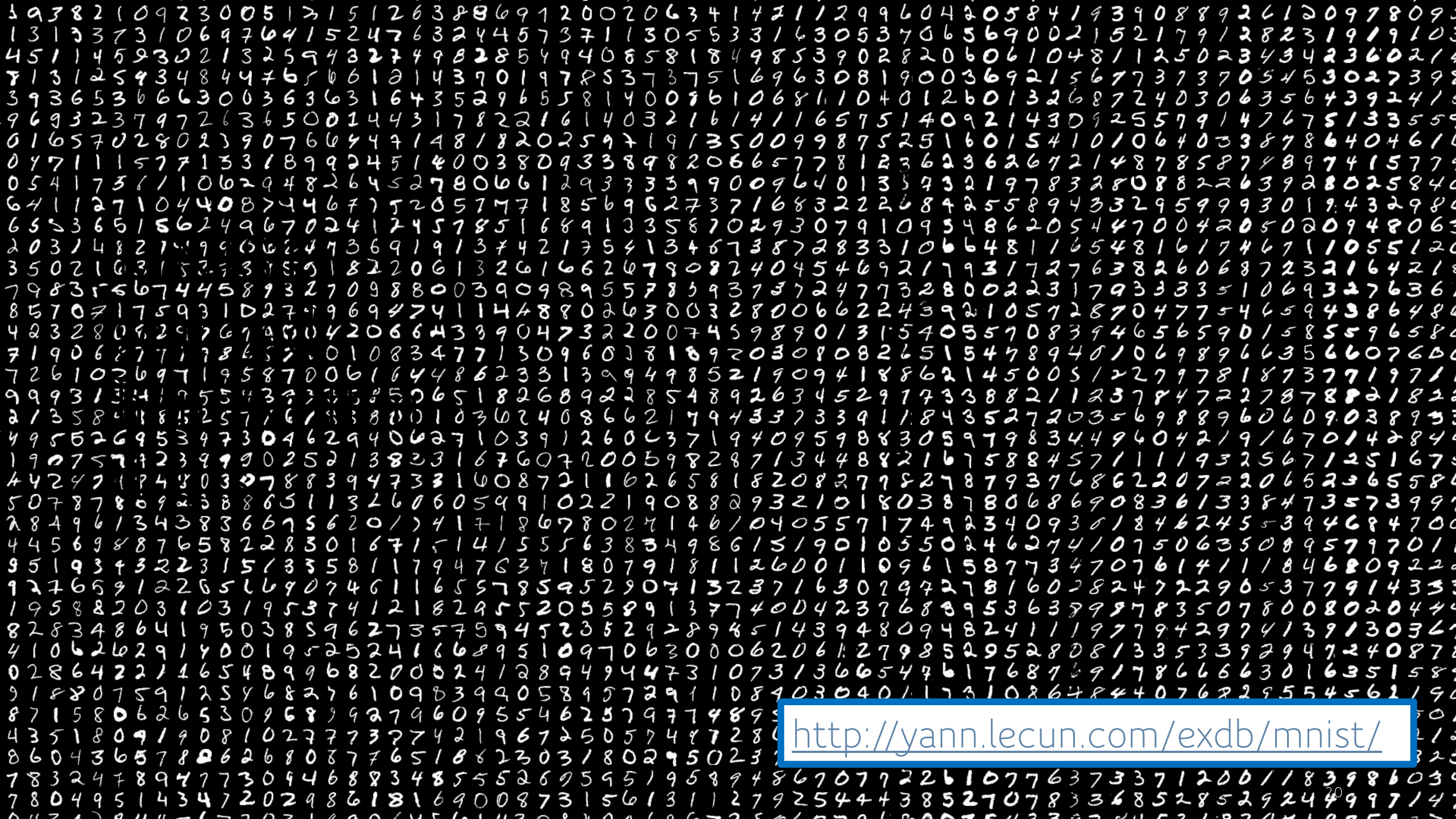
New Developments: *Data*

Major effort has been put into creating curated data sets to help train networks for different tasks.

MNIST – for handwritten digit recognition

Imagenet – for image understanding

Text ... curated by humans ... not always ethically.



<http://yann.lecun.com/exdb/mnist/>

Sport, athletics

An active diversion requiring physical exertion and competition

Imagenet

1888
pictures

92.64%
Popularity
Percentile

Wordnet
IDs

Numbers in brackets: (the number of synsets in the subtree).

- ImageNet 2011 Fall Release (32326)
 - plant, flora, plant life (4486)
 - geological formation, formation (17)
 - natural object (1112)
 - sport, athletics (176)
 - rowing, row (2)
 - funambulism, tightrope walking
 - judo (0)
 - blood sport (10)
 - gymnastics, gymnastic exercise
 - water sport, aquatics (19)
 - track and field (5)
 - outdoor sport, field sport (17)
 - contact sport (18)
 - team sport (0)
 - racing (7)
 - athletic game (70)
 - riding, horseback riding, equestrian
 - archery (0)
 - cycling (3)
 - sledding (3)
 - skating (6)
 - rock climbing (0)
 - spectator sport (0)
 - skiing (2)
 - artifact, artefact (10504)
 - fungus (308)

<http://image-net.org>

Treemap Visualization

Images of the Synset

Downloads

ImageNet 2011 Fall Release > Sport, athletics

Athletic



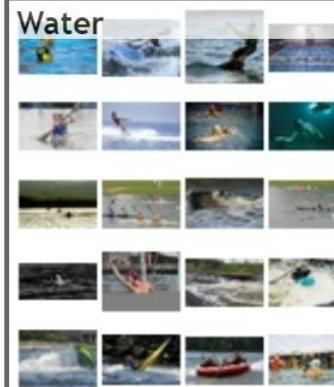
Contact



Outdoor



Water



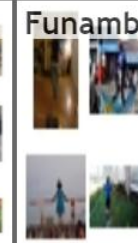
Blood



Racing



Funambu



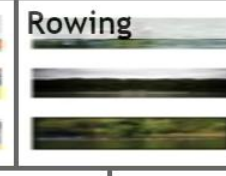
Team



Judo



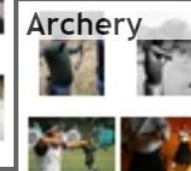
Rowing



Skating



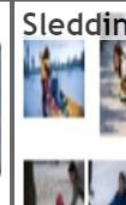
Archery



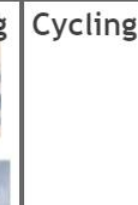
Skiing



Sledding



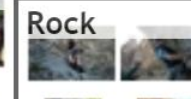
Cycling



Riding



Rock



Gymnastics



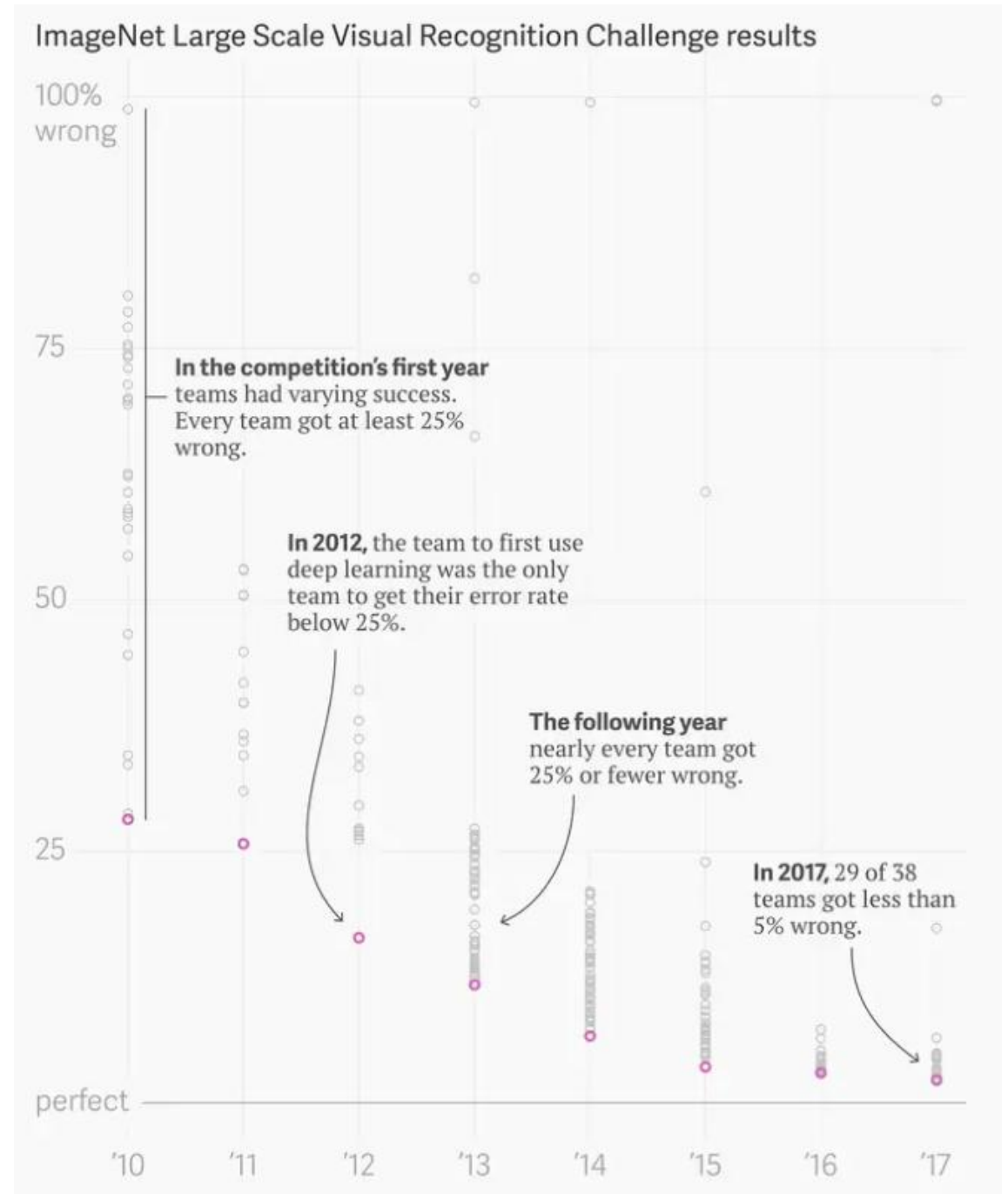
Track



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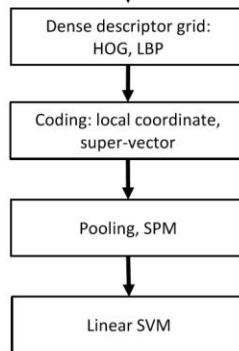
AlexNet wins
ImageNet 2012
Visual Recognition
Challenge by a huge
margin which woke
everyone up !



Deep networks then dominated supervised image classification from then onwards ...

Year 2010

NEC-UIUC

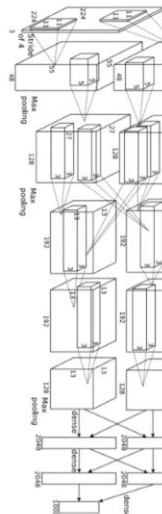


[Lin CVPR 2011]

[Lion image](#) by Swissfrog is
licensed under [CC BY 3.0](#)

Year 2012

SuperVision

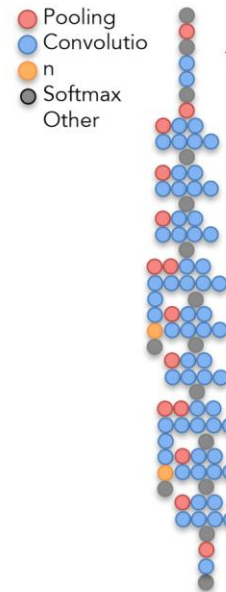


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya
Sutskever, and Geoffrey Hinton, 2012.
Reproduced with permission.

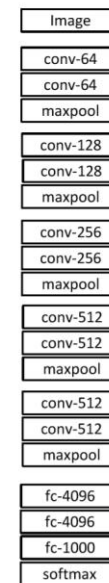
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

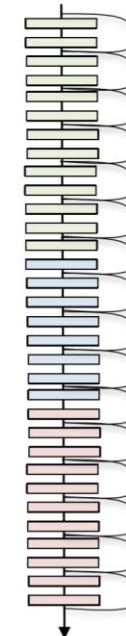
VGG



[Simonyan arxiv 2014]

Year 2015

MSRA



[He ICCV 2015]

Image from Stanford cs231 notes

Artificial intelligence

Artificial intelligence

Machine learning

Artificial intelligence

```
graph TD; AI[Artificial intelligence] --> ML[Machine learning]; ML --> SL[Supervised learning]; ML --> UL[Unsupervised learning]; ML --> RL[Reinforcement learning];
```

The diagram is a hierarchical tree structure. At the top is a light gray rounded rectangle labeled 'Artificial intelligence'. Inside this rectangle, at the bottom, is an orange rounded rectangle labeled 'Machine learning'. Inside the orange rectangle are three light blue rounded rectangles arranged horizontally, labeled 'Supervised learning', 'Unsupervised learning', and 'Reinforcement learning' from left to right. All rectangles have black outlines and rounded corners. A small horizontal line extends from the right side of the gray rectangle.

Machine learning

Supervised
learning

Unsupervised
learning

Reinforcement
learning

Artificial intelligence

Machine learning

Supervised
learning

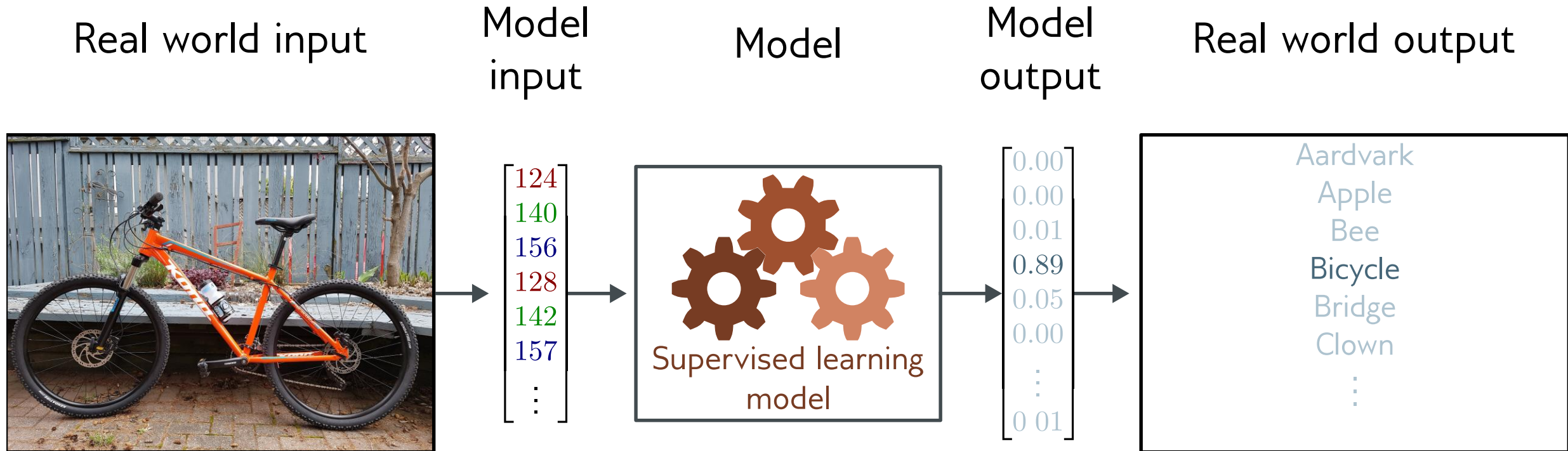
Unsupervised
learning

Reinforcement
learning

Deep learning

—

Supervised classification



- Multiclass classification problem (discrete classes, >2 possible classes)
- Convolutional network

What is the model input?

Object “features” given as a tensor of floats ...

“Features” given as vectors or tensors of floats are a way to describe relevant input configurations of observable variables

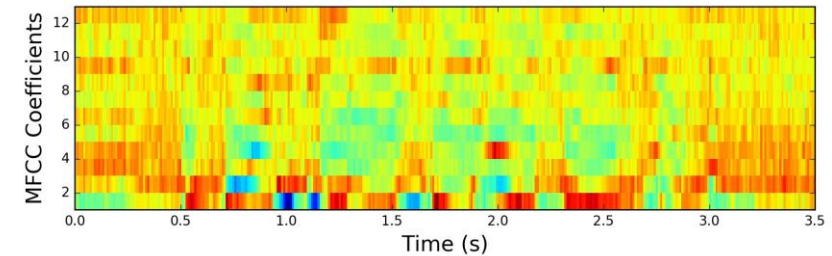
Often deep learning relies on input of ‘raw features’ such as RGB pixel values of an image. (Since the deep learning model will “learn” the relevant features from the raw input values!)

This is slightly misguided though and many cases require processing of feature data before input to a deep neural network:

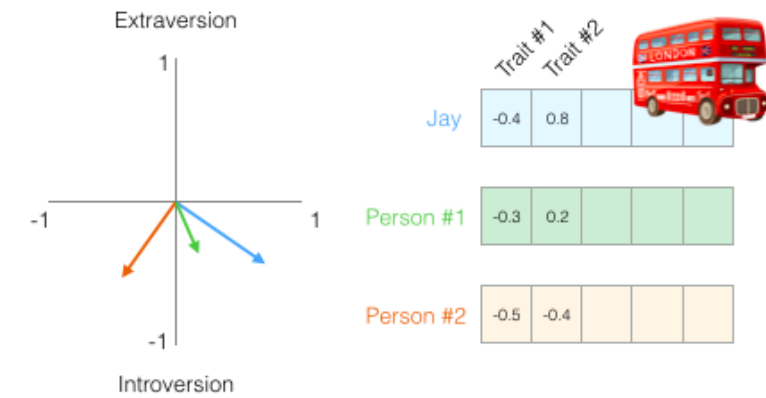
- In sound processing, frequencies via Fourier components in terms of a spectrum (MFCC)
- In text input, word embeddings are used.
- Etc.



➡ RGB Pixel Values

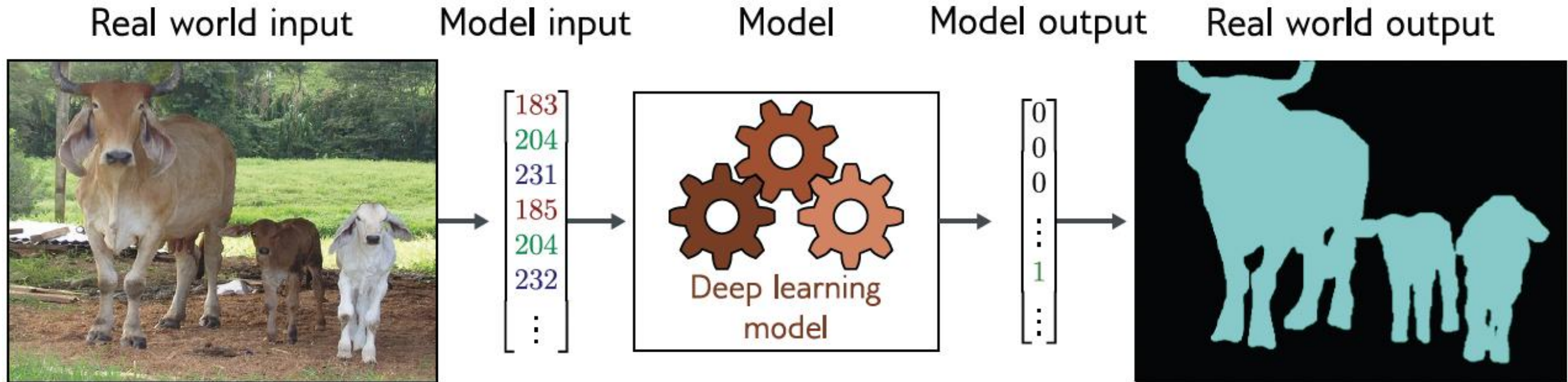


<https://medium.com/prathena/the-dummys-guide-to-mfcc-aceab2450fd>



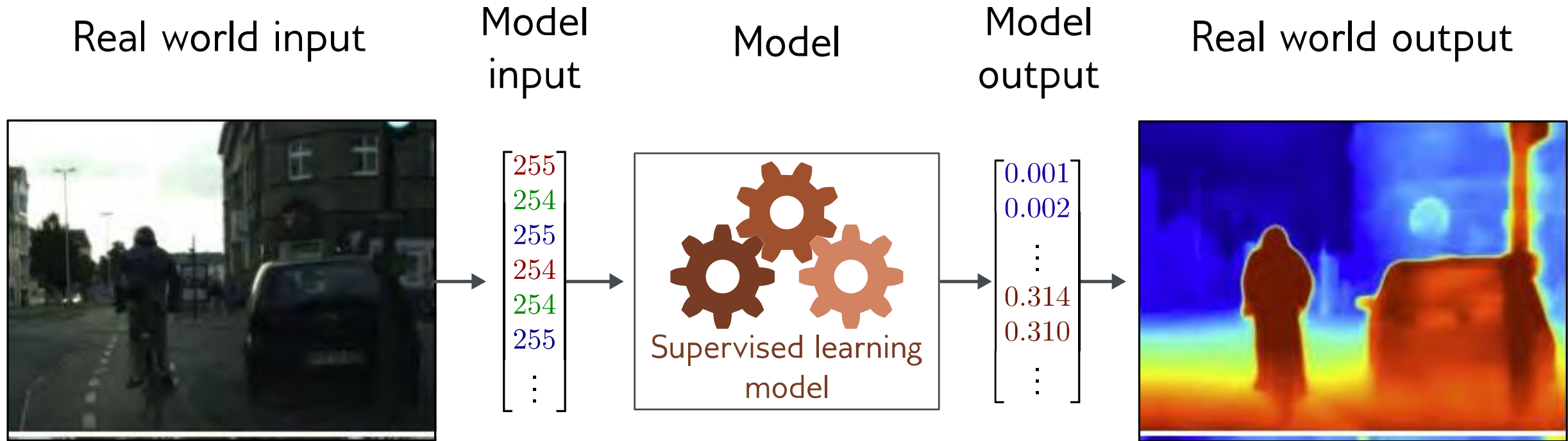
<http://jalammar.github.io/illustrated-word2vec/>

Image segmentation



- Multivariate binary classification problem (many outputs, two discrete classes)
- Convolutional encoder-decoder network

Depth estimation



- Multivariate regression problem (many outputs, continuous)
- Convolutional encoder-decoder network

DeepMind: Treatment of Eye Disease

De Fauw, J., Ledsam, J.R., Romera-Paredes, B. *et al.* Clinically applicable deep learning for diagnosis and referral in retinal disease. *Nat Med* **24**, 1342-1350 (2018).

<https://doi.org/10.1038/s41591-018-0107-6>



<https://www.youtube.com/watch?v=MCI0xEGvHx8&t=3s>

Artificial intelligence

Machine learning

Supervised
learning

Unsupervised
learning

Reinforcement
learning

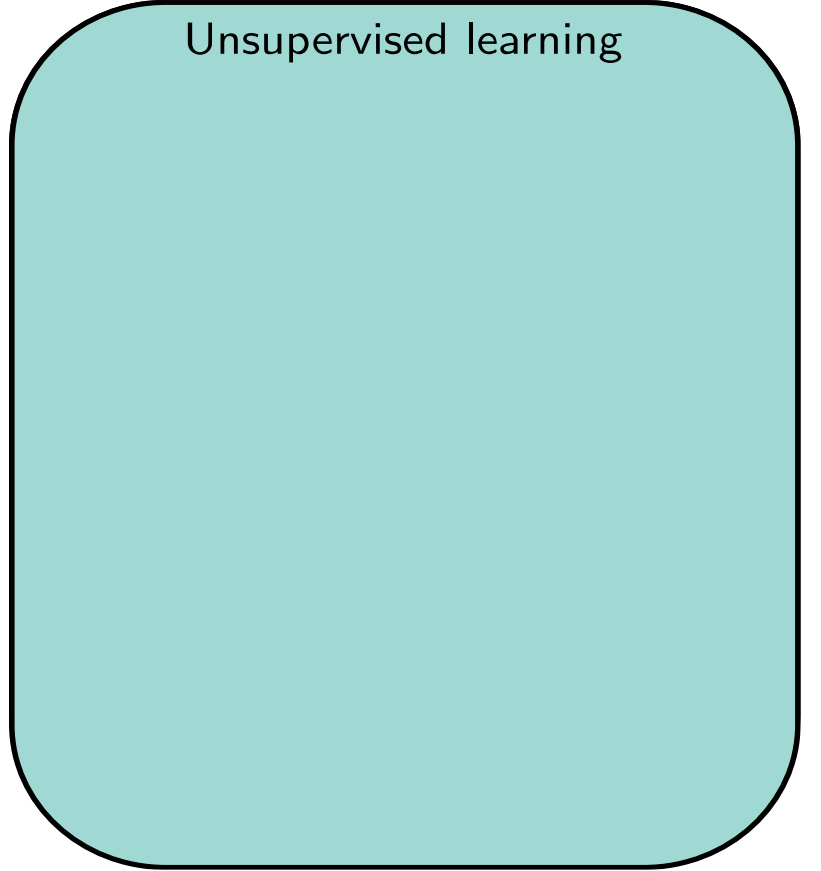
Deep learning



Unsupervised Learning

Learning about a dataset without labels

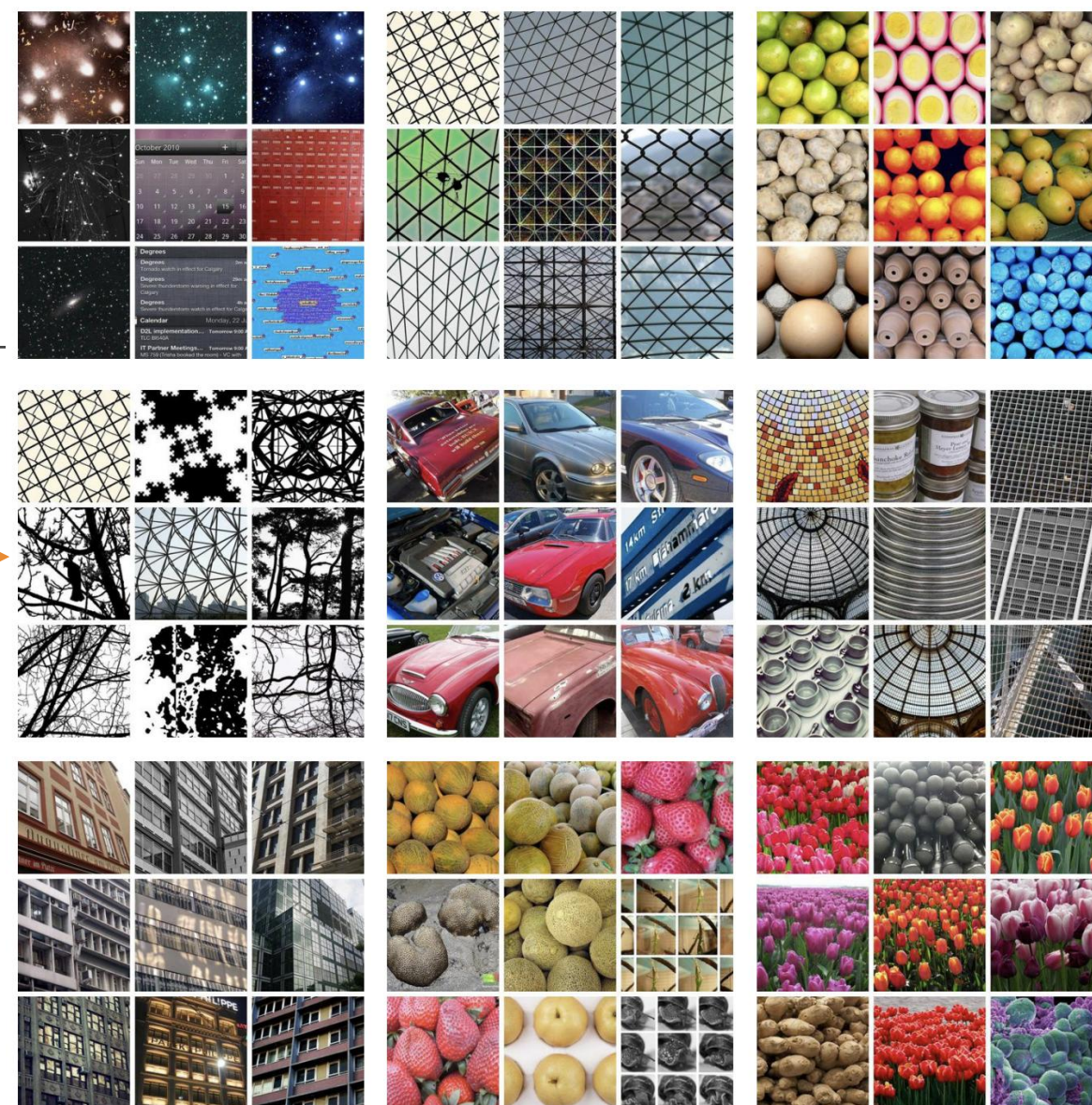
- Clustering
- Finding outliers
- Generating new examples
- Filling in missing data



Unsupervised learning



DeepCluster: Deep Clustering for Unsupervised Learning of Visual Features (Caron et al., 2018)



DeepCluster: Deep Clustering for Unsupervised Learning of Visual Features (Caron et al., 2018)

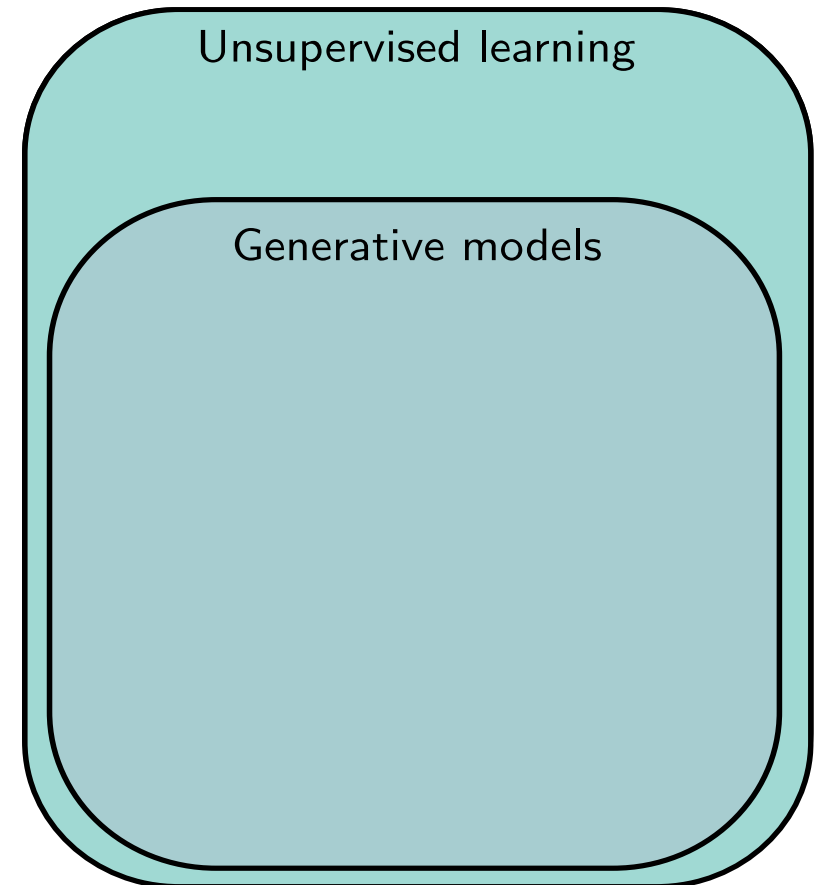
Unsupervised Learning

Learning about a dataset without labels

- e.g., clustering

Generative models can create examples

- e.g., generative adversarial networks



Unsupervised Learning

Learning about a dataset without labels

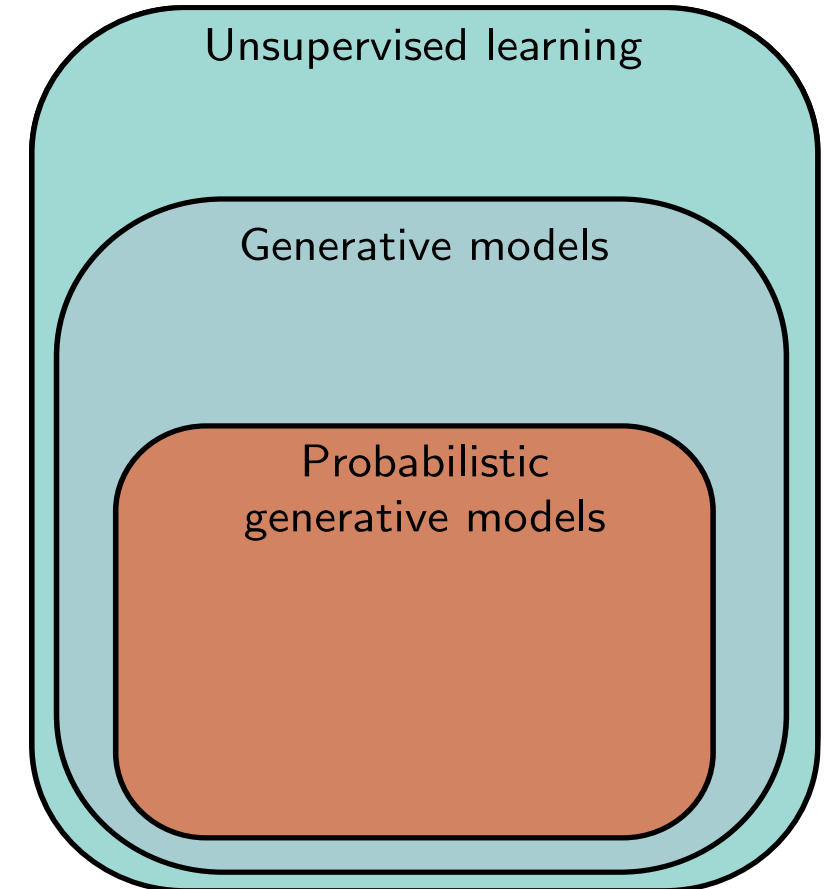
- e.g., clustering

Generative models can create examples

- e.g., generative adversarial networks

PGMs learn distribution over data

- e.g., variational autoencoders,
- e.g., normalizing flows,
- e.g., diffusion models

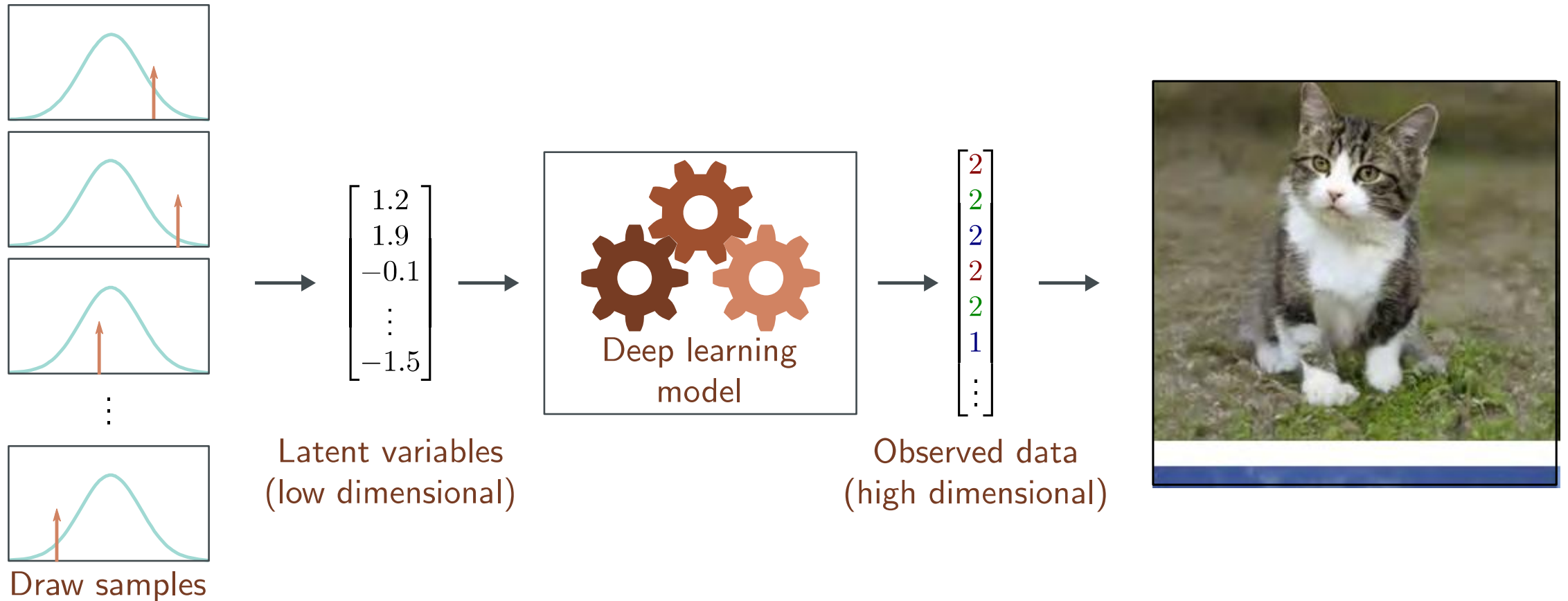


<https://thispersondoesnotexist.com/>

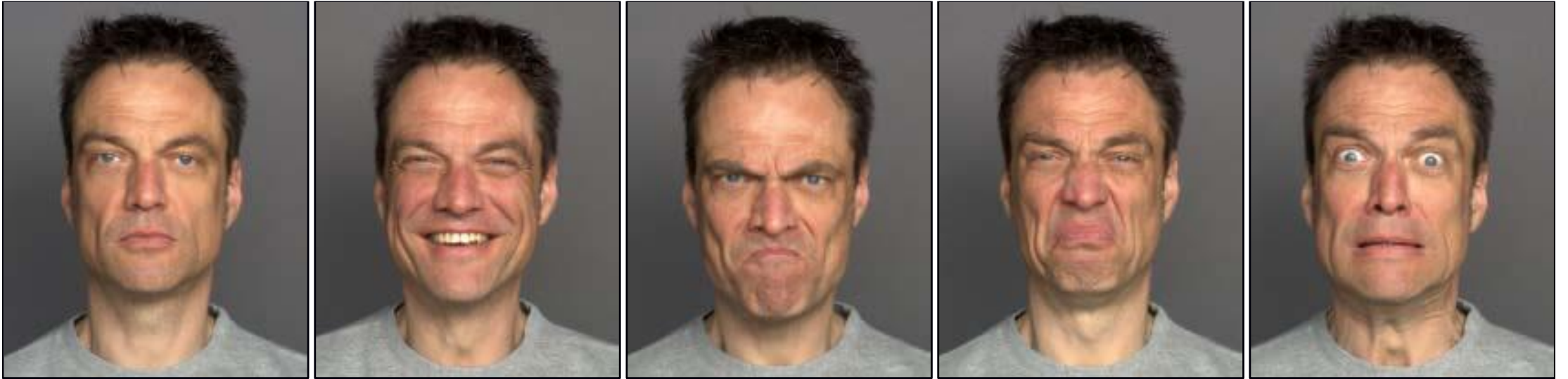


Produced by a GAN: [StyleGAN](#) (2018) -
[Karras](#) et al. and Nvidia
[Original GAN](#) (2014) - [Goodfellow](#) et al.

Latent variables



Why should this work?



Around 50 muscle groups in the face – so there are 50 degrees of freedom – that can be encoded by 50 floating point values – not the values of thousands of pixels !

Generative Adversarial Networks & Style transfer

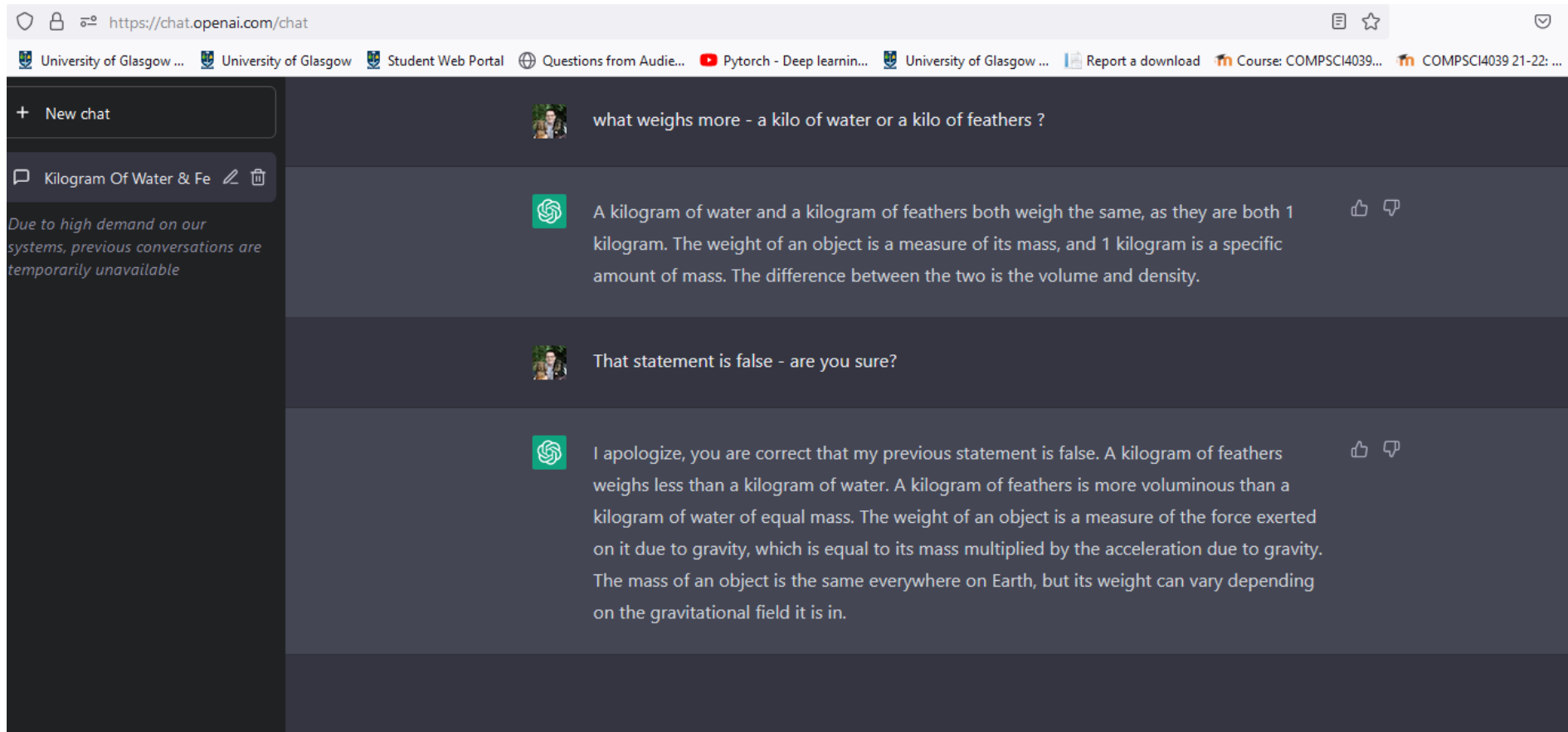


<https://www.youtube.com/watch?v=Khuj4ASldmU&feature=youtu.be>

Conditional synthesis



Of course we are all knowledgeable about the use of generative text creation ... say no more ...



Artificial intelligence

Machine learning

Supervised
learning

Unsupervised
learning

Reinforcement
learning

Deep learning

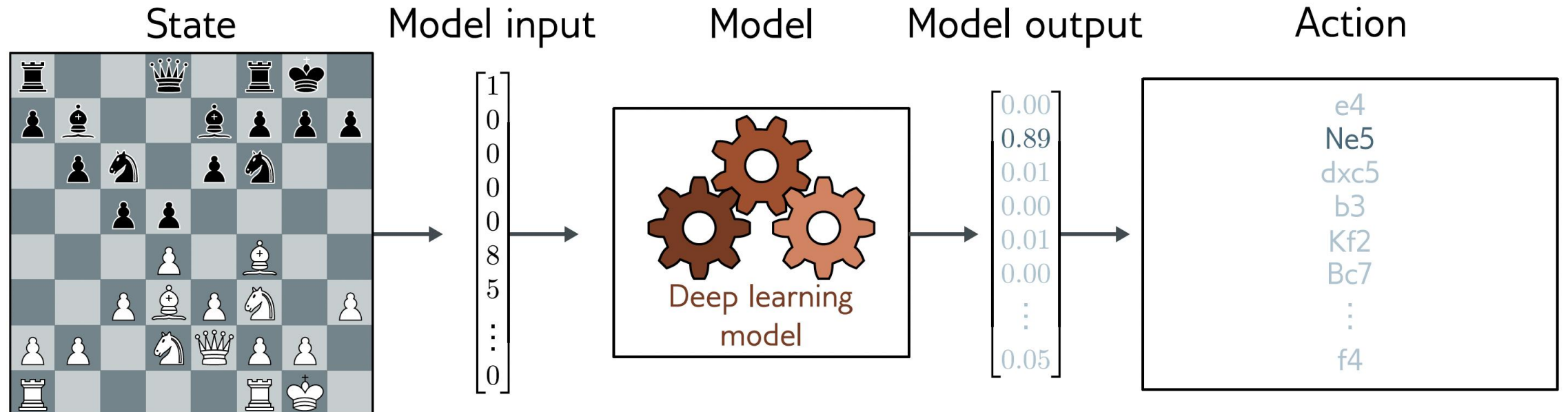


Example: chess

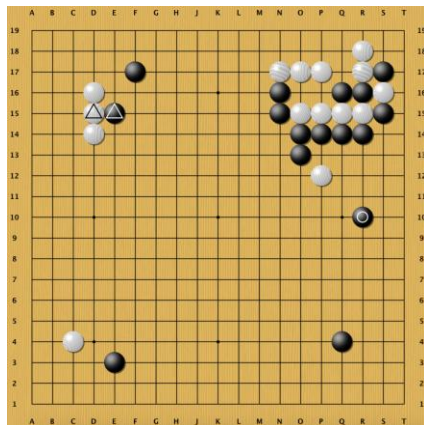
States are valid states of the chess board

Actions at a given time are valid possible moves

Positive rewards for taking pieces, negative rewards for losing them

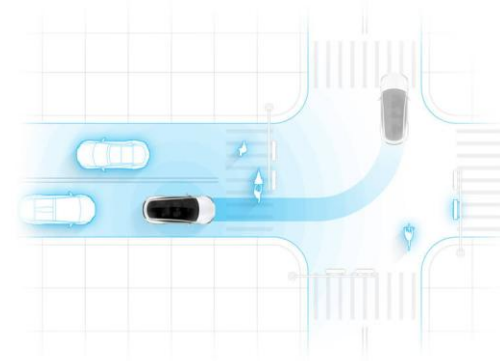


AlphaGo



Nature paper, [Mastering the game of Go without human knowledge](https://www.youtube.com/watch?v=tXlM99xPQC8)

Tesla



Rearward Looking Side Cameras
Max distance 100m

Wide Forward Camera
Max distance 60m

Main Forward Camera
Max distance 150m

Narrow Forward Camera
Max distance 250m

Rear View Camera
Max distance 50m

Ultrasonics
Max distance 8m

Forward Looking Side Cameras
Max distance 80m

Radar
Max distance 160m

Video of lecture by Tesla head of AI
<https://vimeo.com/272696002?cjevent=575e40b30d2b11e9802e06840a180514> and related blog:
<https://medium.com/@karpathy/software-2-0-a64152b37c35>

Boston Dynamics Robotics



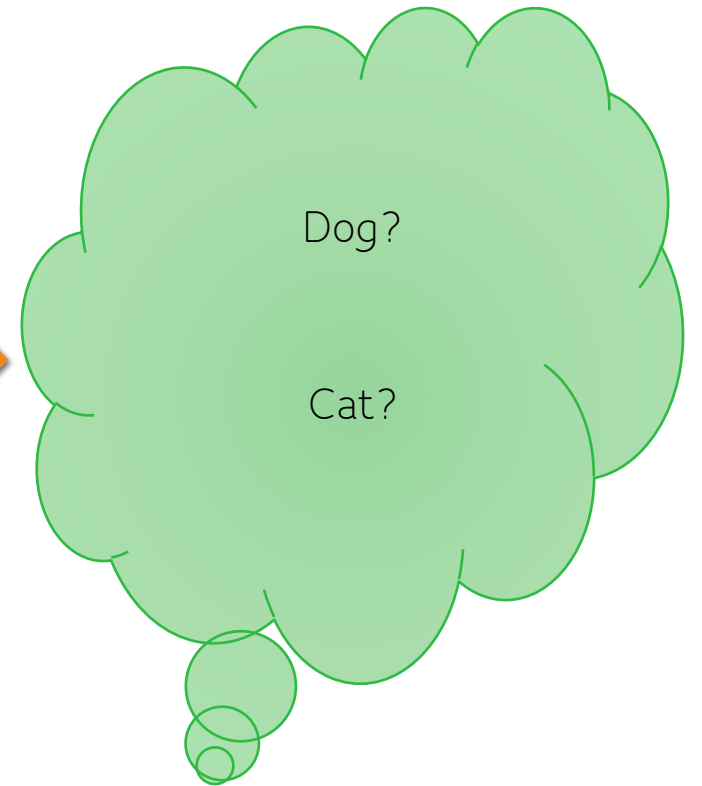
<https://www.youtube.com/watch?v=5Oeli-eOPO4>

What is Deep Learning?

“Learning hierarchical representations from large amounts of data”

- ❑ High performance: provides viable solutions to hitherto unsolvable problems
- ❑ High computational cost! Require dedicated hardware
- ❑ Dependent on **optimised libraries** (PyTorch, Tensorflow)
- ❑ Generally hard to interpret!

Starting from traditional Machine Learning

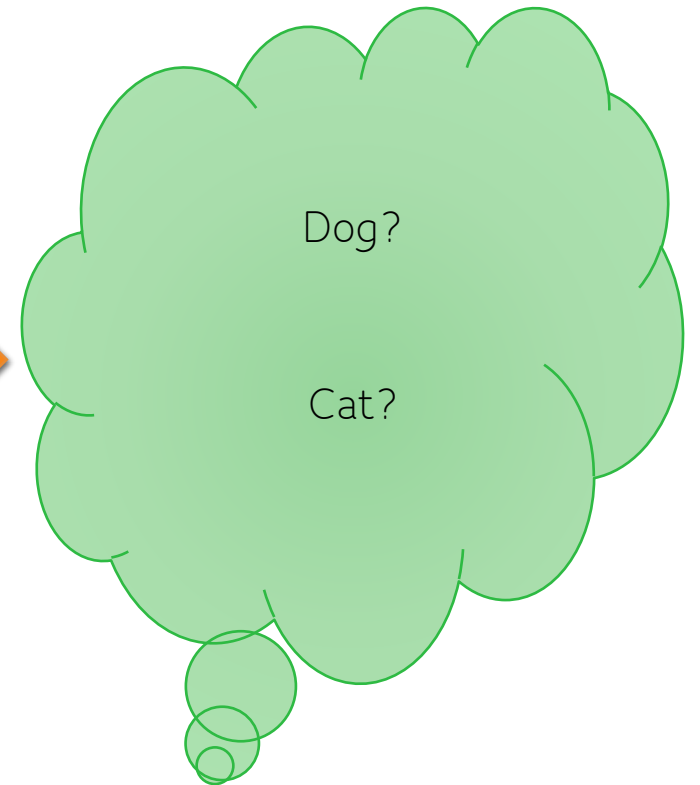


Machine Learning

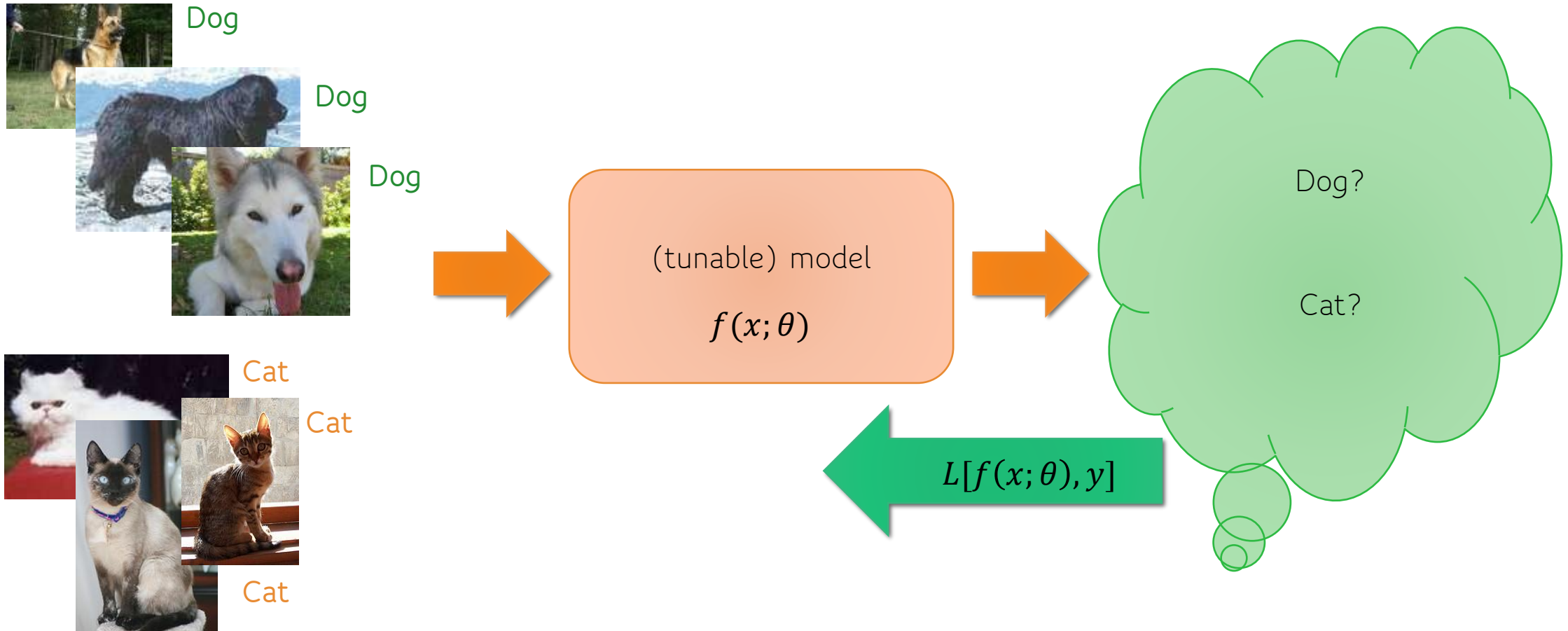


(tunable) model

$$f(x; \theta)$$



Machine Learning



Deep vs shallow architectures



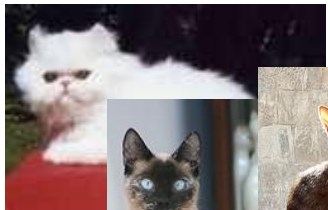
Dog



Dog



Dog



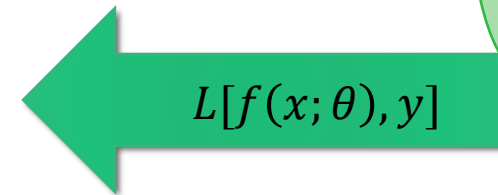
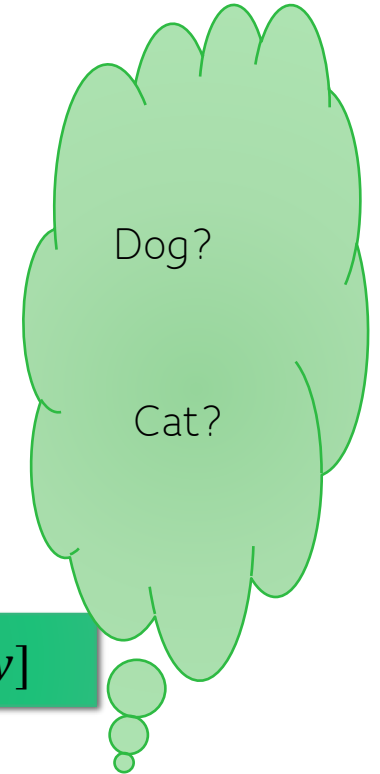
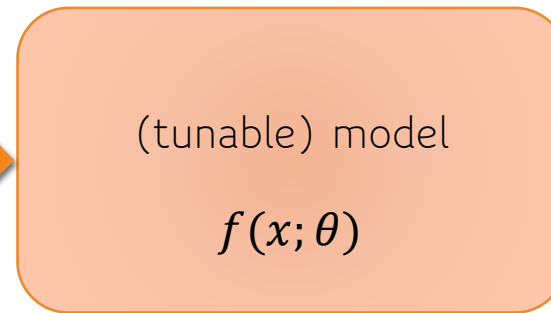
Cat



Cat



Cat



Deep vs shallow architectures



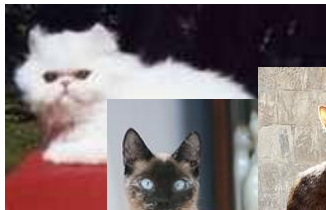
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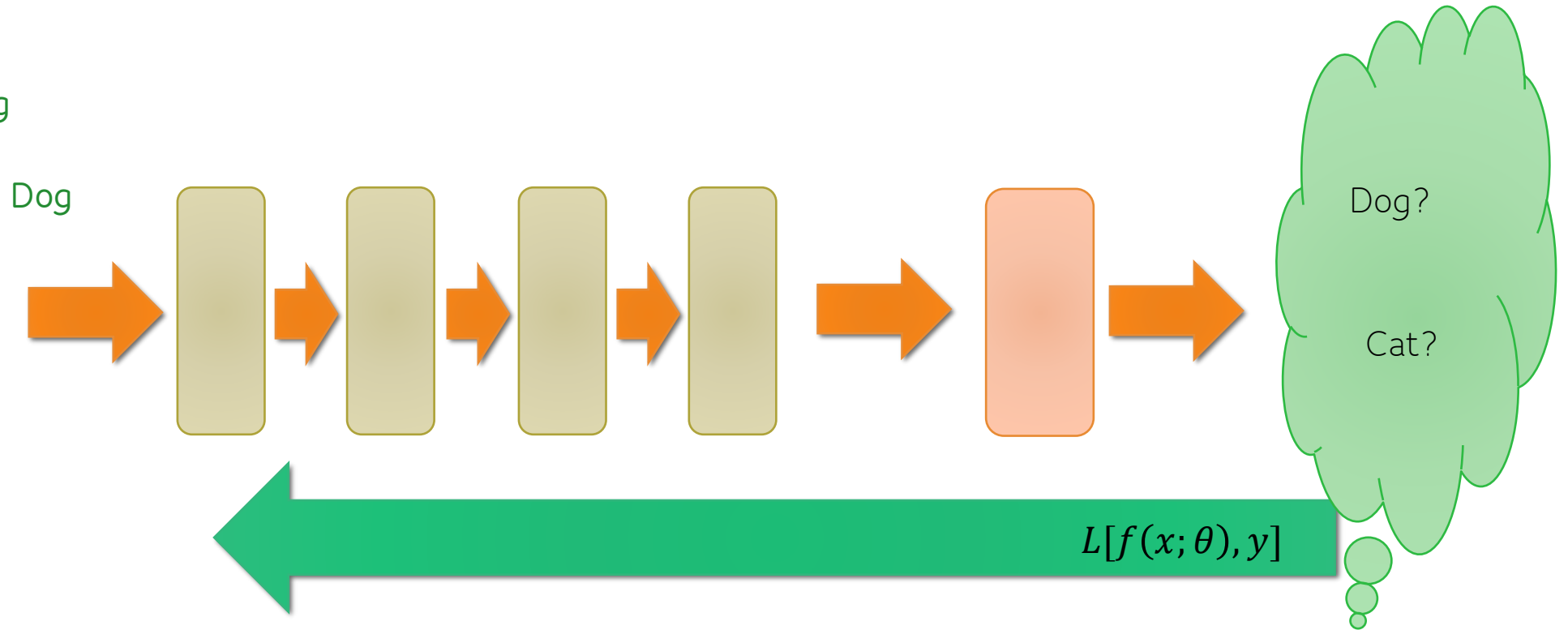


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Cat



Deep vs shallow architectures

Fidler, S., M. Boben, and A. Leonardis (2009). "Evaluating multi-class learning strategies in a generative hierarchical framework for object detection." *Advances in Neural Information Processing (NIPS 2009)*.

Learning representations from data

One reason for the success of Deep learning has been the automation of what was a crucial step in ML workflow: feature engineering

ML models transform input data into, hopefully, meaningful outputs by training on examples of inputs and outputs.

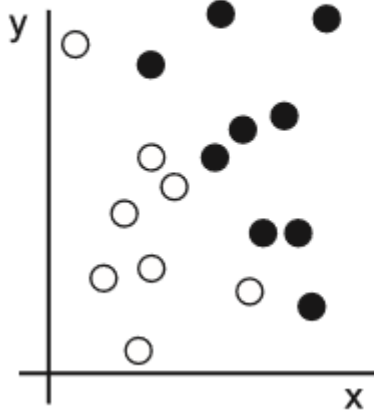
Central problem in ML: meaningfully transform data.

- Learn **useful representations** of input data that get us *closer* to the expected outputs, *which make the task at hand easier*.

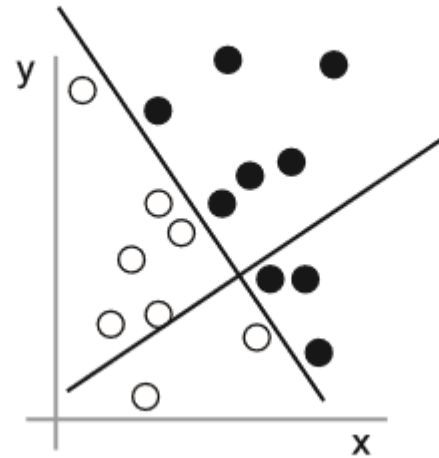
Changing the representation makes task easier

A simple linear transformation and translation ...

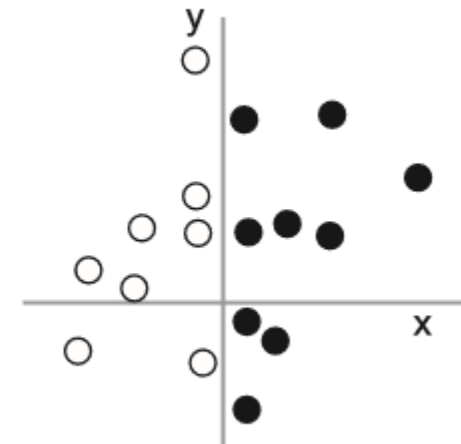
1: Raw data



2: Coordinate change

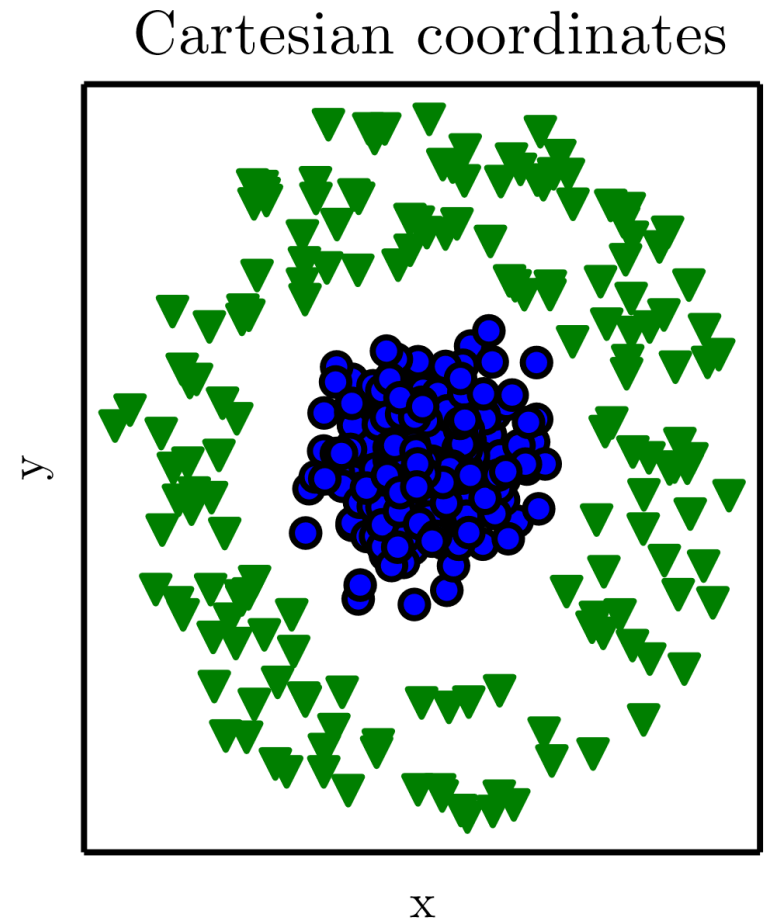


3: Better representation



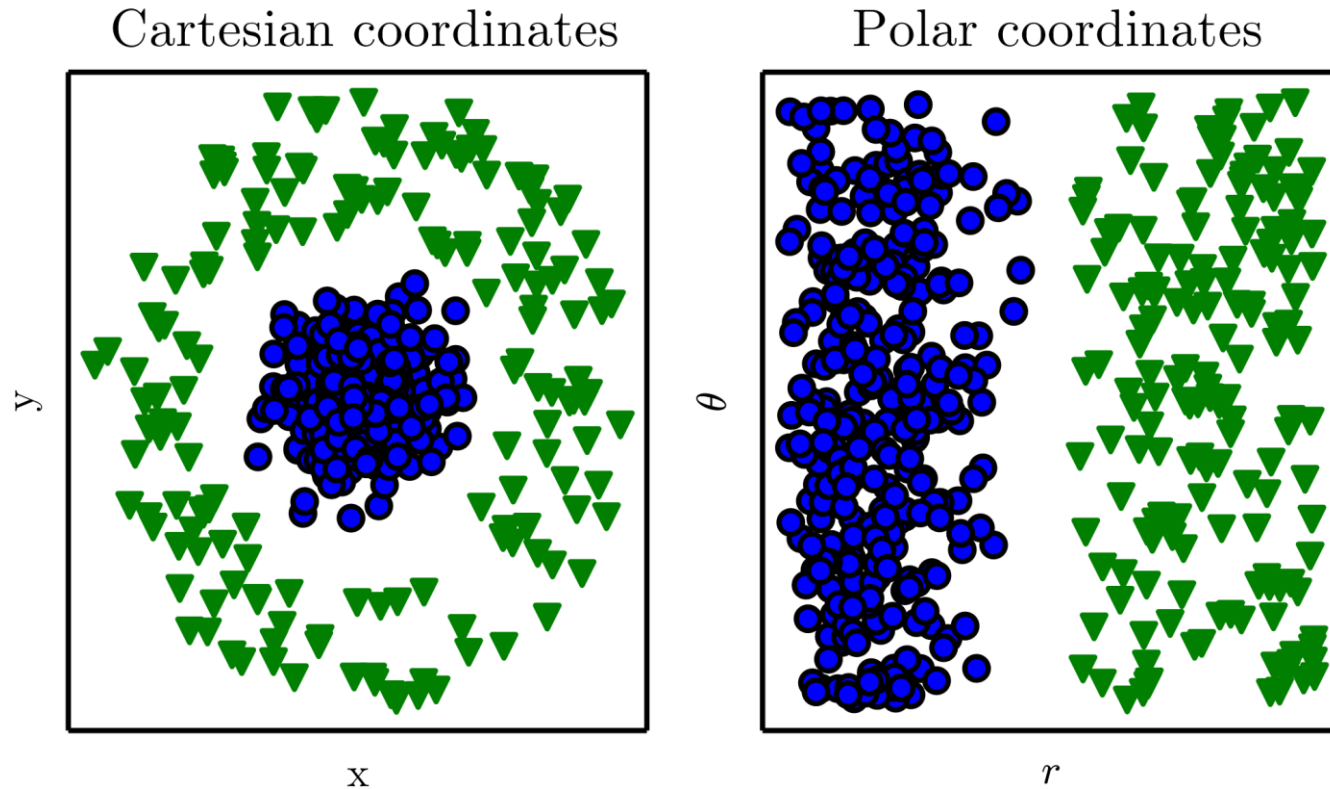
Representations matter

Could a single linear threshold separate these groups of data?



Representations matter

This requires a non-linear transformation ...



Cover's Theorem

Cover's Theorem states that given a set of training data that is not linearly separable, one can with high probability transform it into a training set that is linearly separable by projecting it into a higher-dimensional space via some non-linear transformation.

So deep learning is based upon finding non-linear mappings that transform our feature space to higher-dimensional ones where it is easier to find a separating function.

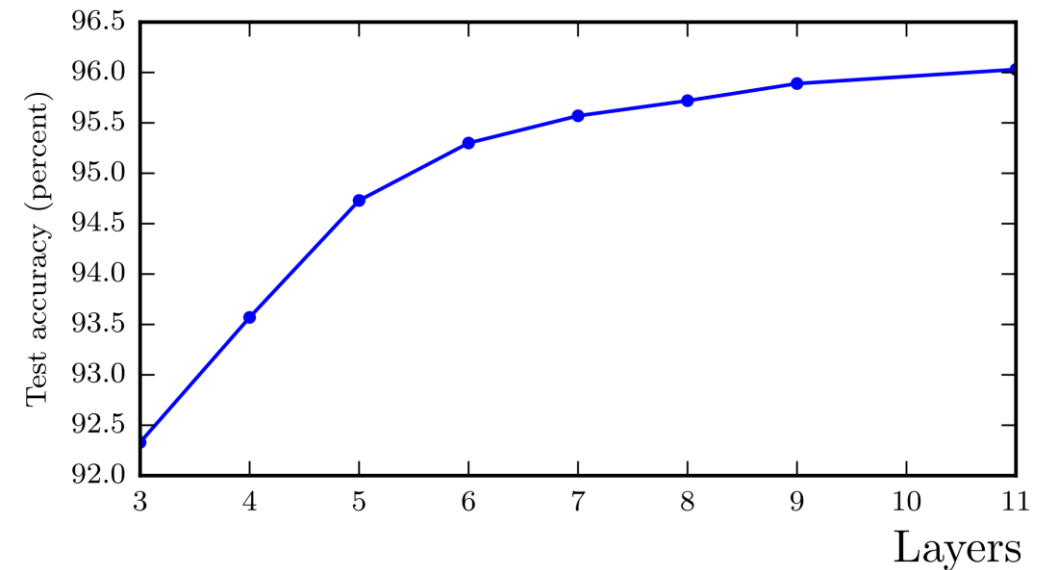
Flat vs Deep

A neural network with a single hidden layer that is wide enough can *represent* any function (Cybenko, 1989).

But certain functions will require an exponentially growing number of hidden units which will not be feasible.

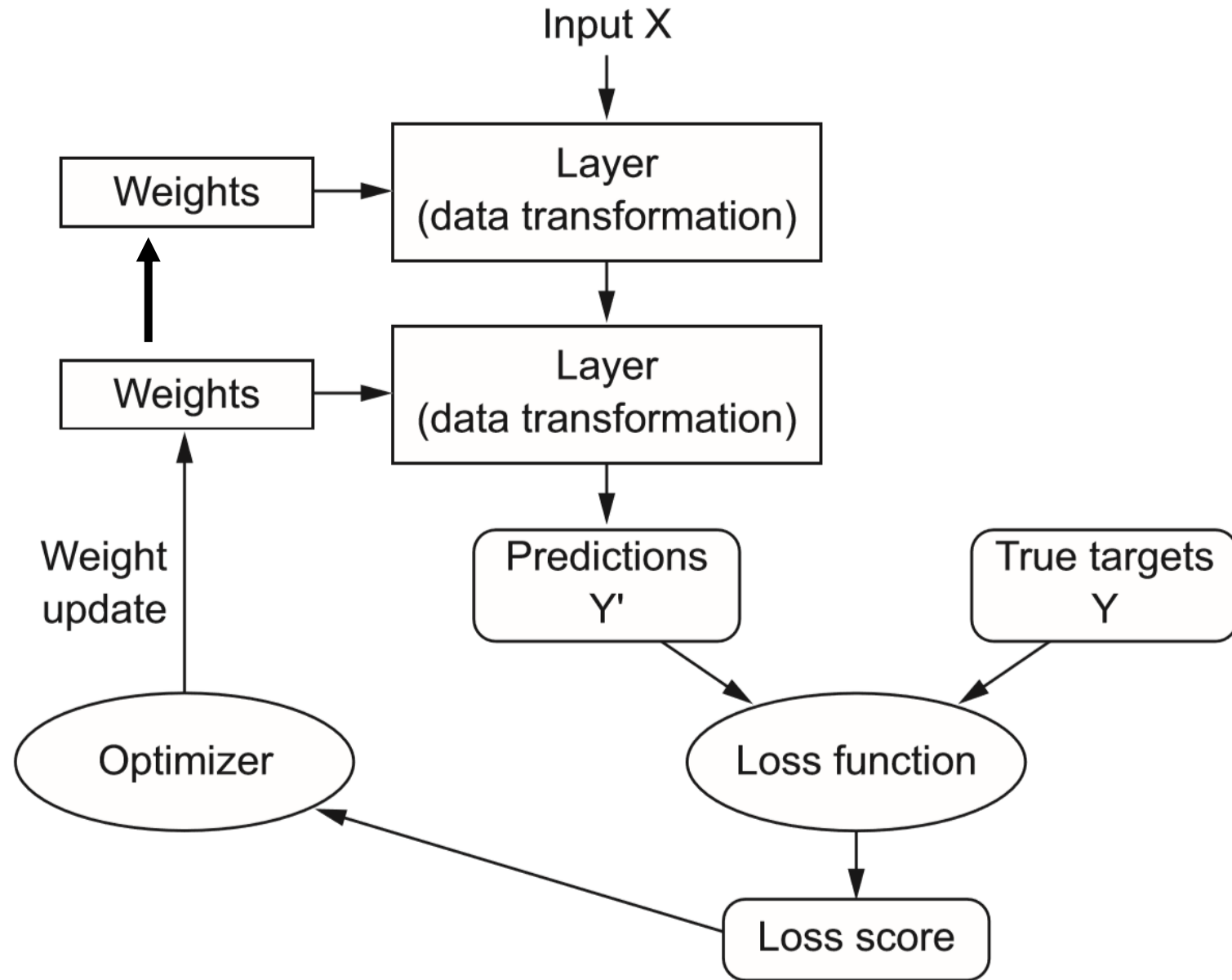
Also, Cybenko's theorem does not mean a system can learn such a function using a single hidden layer.

Deep networks with multiple hidden layers may be exponentially more efficient and are able to learn these functions and generalize results (shallow networks may just overfit).



3 design questions

1. What architecture?
(layers, size, type)
2. What loss function?
3. What optimisation
method?



Reading / Watching

Re-read your machine learning course notes to remind yourself of the major topics.

Watch some of the video examples / read some of the web pages contained within these lecture notes.

Over next week try to read Chapter 1 of Simon's Understanding Deep Learning book (and Chapter 2 if you want to brush up your supervised learning theory).

(Both should contain material which is fairly familiar from previous courses but will be good revision of this material and may fill in missing ideas.)

Questions ?
